Multi-Graph Contextual Learning for Multimodal Meme Classification

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Abstract

This work proposes a **multi-graph contextual classification framework for multimodal meme analysis** that leverages relationships between samples rather than treating them independently. Experiments are conducted on the **What's Beneath Misogynous Stereotyping (WBMS) Dataset**, which contains meme images and captions annotated for misogynistic content. Text and image features are extracted using **BERT** and **CLIP**, respectively, and reduced using Principal Component Analysis (**PCA**). Three similarity graphs are then constructed using cosine similarity–based k-nearest neighbors (**kNN**): a text semantic graph, a visual semantic graph, and a class-discriminative graph obtained using Linear Discriminant Analysis (**LDA**). These graphs are processed using Graph Attention Networks (**GAT**) to learn contextual node representations through attention-based neighbor aggregation. The contextual features from the graphs are fused to perform final classification. This approach aims to incorporate dataset-level contextual information by modeling statistical relationships between samples in the embedding space.

Keywords: Multimodal Meme Classification, Graph Neural Networks, Graph Attention Networks

Literature Review

The reviewed papers explore different approaches combining **k-Nearest Neighbors (kNN)** and **Graph Neural Networks (GNNs)**. This review was conducted to understand how neighbor retrieval, graph structures, and semantic relationships can be integrated for building graphs of embeddings and class semantics. These studies informed the design of our approach, where a graph is constructed using cosine similarity between embeddings, and contextual node representations are learned using graph convolution with **kNN-based neighborhoods** and **attention-based weighting**.

1. KNN-GNN: Enhancing GNNs via Common Subspace

This model addresses the limitation of traditional GNNs that rely on the homophily assumption, where connected nodes are expected to share similar labels.

- **Mechanism**: The model projects nodes into a shared latent subspace where similarity is determined by feature representations rather than by predefined graph edges.
- **Architecture**: The framework follows a three-stage process: learning local node representations through attention-based aggregation from 1-hop neighbors, identifying non-local k-nearest neighbors in the learned subspace using K-D Trees, and combining both local and non-local information to form the final node embedding.
- **Outcome**: The approach achieves O(NlogN) complexity and improves performance by enabling nodes to incorporate information from feature-similar nodes even when they are not directly connected in the original graph.

2. KNNGNN: Learning kNN Hyperparameters via GNNs

This work reformulates the kNN algorithm within a graph neural network framework, allowing the model to learn similarity and weighting functions automatically.

- **Mechanism**: For each query instance, a micro-graph is constructed containing the query node and its k nearest neighbors.
- **Architecture**: Nodes represent the query and neighbor instances, while edges encode feature differences between nodes. The graph is processed using a Message Passing Neural Network (MPNN), where the message and update functions implicitly learn the distance metric and neighbor influence.
- **Outcome**: This approach eliminates the need for manually tuning hyperparameters such as k, distance functions, or weighting schemes, demonstrating how local neighborhood graphs can model contextual relationships during prediction.

3. kNN-KGE: Retrieval-Augmented Knowledge Graph Embedding

This research introduces a semi-parametric knowledge graph embedding framework that improves reasoning for rare or emerging entities.

- **Mechanism:** The model combines a Pre-trained Language Model (PLM) with an external knowledge store containing entity descriptions and relational triples.
- **Architecture:** During inference, link prediction is treated as a Masked Entity Modeling task. The representation of the masked token is used to retrieve k nearest entities from the knowledge store, and the retrieved information is combined with the model's predictions.
- **Outcome:** The method improves performance for long-tail entities and remains effective even with reduced training data, illustrating how embedding retrieval from external memory can enhance contextual reasoning.

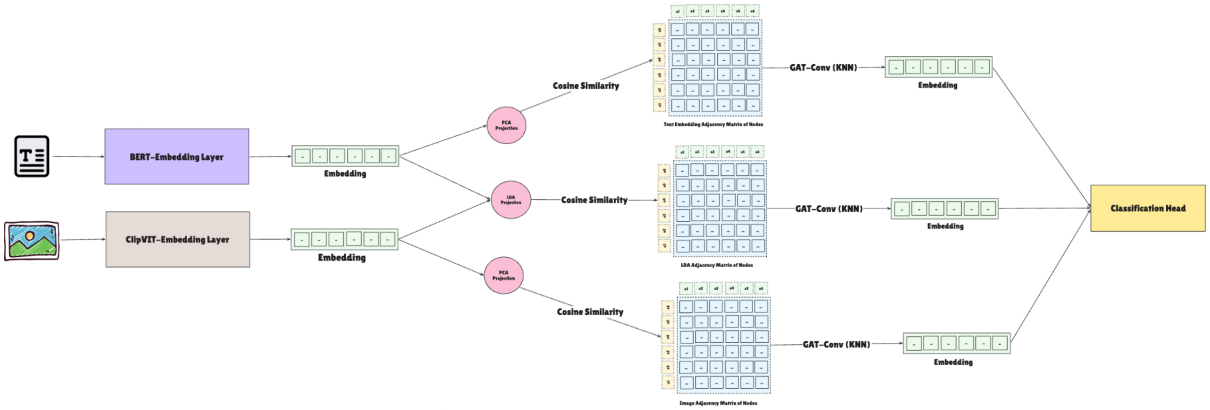
Proposed Methodology

The proposed method introduces a multi-graph contextual classification framework that models relationships between samples in a multimodal dataset. Instead of predicting from independent features, the dataset is represented as a set of similarity graphs constructed using cosine similarity–based k-nearest neighbor (kNN) connections.

Three complementary graphs are constructed:

1. Text Semantic Graph
2. Visual Semantic Graph
3. Class-Discriminative Graph

Each graph captures a different statistical structure of the dataset. The graphs are processed using Graph Attention Networks (GAT) to learn contextual node representations, and the resulting features are fused for final classification.



Multimodal Feature Extraction

Each data sample contains textual and visual components.

Text captions are encoded using BERT to obtain semantic text embeddings, while images are encoded using CLIP to generate visual feature representations. Since these embeddings are high-dimensional, Principal Component Analysis (PCA) is applied separately to text and image features to obtain compact representations suitable for graph construction.

Graph Construction

Three graphs are constructed to model different forms of relationships among samples.

- **Text Semantic Graph:** The text semantic graph captures linguistic relationships between captions. A kNN graph is constructed using cosine similarity between PCA-reduced BERT embeddings, connecting each caption to semantically similar captions in the dataset.
- **Visual Semantic Graph:** The visual semantic graph represents similarity between images. Cosine similarity is computed over PCA-reduced CLIP image embeddings, and each image node is connected to its nearest visual neighbors.
- **Class-Discriminative Graph:** To model class-level statistical relationships, text and image PCA features are combined to form a multimodal representation. Linear Discriminant Analysis (LDA) is then applied to project the data into a space that maximizes class separability. A kNN graph is constructed in this LDA space, connecting samples that are close in terms of discriminative class structure.

Contextual Representation Learning

Each graph is processed using a Graph Attention Network, which allows nodes to aggregate information from neighboring samples while learning attention weights that determine the importance of each neighbor. This enables the model to refine node representations using contextual information from similar samples.

Feature Fusion and Classification

The contextual representations obtained from the three graphs are combined to form a unified multimodal representation. This fused feature vector is then passed through a classification layer to produce the final prediction. During inference, new samples are connected to the graph using kNN relationships, allowing the model to utilize dataset-level contextual information for classification.

Preliminary Results

Preliminary experiments were conducted to evaluate the effect of incorporating class-semantic relationships before implementing the full multi-graph architecture. At this stage, the modality-specific graphs (text and visual graphs) were not included. Instead, the focus was on analyzing the impact of the class-semantic graph constructed in the LDA space.

Table 1: Performance Comparison of Models

Model	Split	Accuracy	Precision	Recall	F1 Macro	F1 Weighted
Graph	Train	0.9847	0.9847	0.9847	0.9709	0.9846
	Test	0.9554	0.9550	0.9554	0.9247	0.9543
No Graph	Train	0.9425	0.9410	0.9425	0.8869	0.9395
	Test	0.9061	0.9038	0.9061	0.8144	0.8977
PCA Only	Train	0.9982	0.9983	0.9982	0.9968	0.9982
	Test	0.9507	0.9490	0.9507	0.9099	0.9491

Three model variations were evaluated:

1. **PCA Only**: Multimodal embeddings were reduced using Principal Component Analysis (PCA) and directly used for classification.
2. **PCA + LDA**: PCA-reduced features were projected using Linear Discriminant Analysis (LDA) to capture class-discriminative structure. The PCA features were concatenated with the LDA-projected features to form the final representation used for classification.
3. **PCA + LDA + Graph**: A k-nearest neighbor graph was constructed using cosine similarity in the LDA space, allowing samples to aggregate information from class-related neighbors before classification.

The results indicate that incorporating class-semantic neighborhood information improves the discriminative capability of the model compared to using feature representations alone. These findings support the hypothesis that modeling relationships between samples can enhance classification performance. Future work will extend this setup by incorporating the text and visual semantic graphs to enable the full multi-graph contextual learning framework.

Reference Links:

- Literature Review Summary: kNN-KGE (Wang et al., 2023)
- Learning on Multimodal Graphs: A Survey
- KNN-GNN: A powerful graph neural network enhanced by aggregating K-nearest neighbors in common subspace
- k -Nearest Neighbor Learning with Graph Neural Networks
- https://arxiv.org/pdf/2508.03732

GitHub Repository: https://github.com/vishnurchityala/graph-networks/tree/gnn